

Ex: Over $\mathbb{Z}_7$, $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

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Note: We identify $R^e = M_{e \times 1}(R)$

Defn A square matrix $A \in M_{\text{dxd}}(R)$ is invertible (over R) if there exists $B \in M_{\text{dxd}}(R)$ s.t. $AB = BA = I_d = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$.

If it exists, B is uniquely determined of A . We then call $B =: A^{-1}$ the inverse of A (over R).

Q:

- How can we find out if A is invertible?
- If A^{-1} exists, how can we find it?
- Why should we care?

Defn $R^\times := \{a \in R : \exists b \in R. ab = ba = 1 \text{ in } R\}$
is called the unit group of R

(More about "groups": later!)

Ex: $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$

• $\mathbb{Z}^\times = \{1, -1\}$

• $\mathbb{Z}_n^\times = \{a \in \mathbb{Z}_n : \gcd(a, n) = 1\}$

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Thm Let R be a ring with identity and commutative multiplication.

Then a square matrix A over R is invertible over R

if and only if $\det(A) \in R^\times$.

(Hence, the determinant determines invertibility.)

Ex • $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ is not invertible over $\mathbb{Z}_6$ because

$$\det(A) = 2 \notin \mathbb{Z}_6^\times = \{1, 5\}$$

• However, A is invertible over $\mathbb{Z}_3$ because $2 \in \mathbb{Z}_3^\times = \{1, 2\}$.

Indeed, $A^2 = \begin{bmatrix} 4 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ over $\mathbb{Z}_3$.

Defn $GL_d(R) := \{A \in M_{d \times d}(R) : A \text{ is invertible over } R\}$

is called the general linear group of degree d over R .

Higher-dimensional affine ciphers

Fix integers $d \geq 1$ and $n \geq 1$.

Plaintexts = ciphertexts = elements of $\mathbb{Z}_n^d$, regarded as column vectors

Keys: pairs (A, b) , where $A \in \text{GL}_d(\mathbb{Z}_n)$ and $b \in \mathbb{Z}_n^d$

Encryption: $e_{(A,b)}(x) = Ax + b \in \mathbb{Z}_n^d$ for $x \in \mathbb{Z}_n^d$

Decryption: $d_{(A,b)}(y) = e_{(B, -Bb)}(y)$, where $y \in \mathbb{Z}_n^d$

and B is the (uniquely determined) inverse of A over $\mathbb{Z}_n$.

Remark We won't discuss the problem of finding inverses of matrices over general rings, say with commutative multiplication.

One approach uses $\rightarrow$ adjugate matrices. Over $\mathbb{Z}_n$, we can

also use Gaussian elimination to find an inverse over $\mathbb{Q}$,

then reduce it mod n .

Ex let $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ over $\mathbb{Z}_3$. Over $\mathbb{Q}$, we find that

$$A^{-1} = \begin{bmatrix} 1/2 & -1/2 \\ 0 & 1 \end{bmatrix} \text{ which reduces to } \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \text{ mod } 3.$$

Here, the reduction modulo n of a rational number 28

$x = \frac{a}{b}$ with $\gcd(b, n) = 1$ is defined via

$$x \bmod n := (a \bmod n) \cdot \bar{b} \bmod n,$$

where $\bar{b} \in \mathbb{Z}_n^\times$ is the inverse of $b \bmod n$ in $\mathbb{Z}_n^\times$.

Propn For (A, b) as above and all $x \in \mathbb{Z}_n^d$, we have

$$d_{(A, b)}(e_{(A, b)}(x)) = x.$$

Hence, we obtain a cryptosystem, called a general or, if $d > 1$, higher-dimensional affine cipher.

Pf: Similar to the 1-dimensional case. //

Note that we recover the previous "affine cipher" for $d=1$.

Ex let $n=3$, $d=2$, $A = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ (so $B=A$), $b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\text{Plaintext } \begin{bmatrix} 0 \\ 2 \end{bmatrix} \leadsto \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$$\text{Plaintext } \begin{bmatrix} 0 \\ 1 \end{bmatrix} \leadsto \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Special cases of affine ciphers

- Ordinary affine ciphers are the case $d=1$. For $d=1$ and $A = [1]$, we obtain shift ciphers.
- For $A = I_d = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$ and $b \in \mathbb{Z}_n^d$ arbitrary, encryption takes the form $x \mapsto x + b \pmod{n}$ and decryption is $y \mapsto y - b \pmod{n}$... this is a Vigenère cipher.
- For $A \in GL_d(\mathbb{Z}_n)$ and $b = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$, encryption is $x \mapsto Ax$ and decryption is $y \mapsto Bx$, where B is the inverse of A over $\mathbb{Z}_n$. This is called a Hill cipher (L.S. Hill, 1929), an early example of a "block cipher". Hill received a patent for a "Message Protector", a mechanical implementation of a 6×6 Hill cipher in 1932.

Higher-dimensional affine ciphers are insecure: They are vulnerable to frequency analysis of blocks, among other weaknesses. $\rightarrow$ Assignment 2